Prices/Changes/Cancellations/Out of Stock

Prices

① Made-to-order products, semi-custom orders, and semi-custom order J Series products are estimated separately. Contact your dealer.
② Since the prices are per unit (excluding SRS), the total unit price of miters and the like is the set price.
③ Product prices may be changed without prior notice.

Changes and Cancellations

[Standard Machined Products *Note 1, Made to Order Products, Semi-custom Products, Semi-custom J Series Products]
Because production begins upon ordering, cancellations and changes cannot be made.

[KHK Quick-Mod Products, Custom Gears]
Because production begins upon ordering, cancellations and changes are difficult.
Changes may be possible in some cases, depending on modification status, but a further estimate will be required. Contact your dealer for details.
As the product is manufactured to customer-specified dimensions, it cannot be used elsewhere; therefore, returns are not possible.

Out of Stock

[Standard Gears (gears in stock), Standard Machined Products *Note 1, KHK Quick-Mod Gears]
Production takes place according to stock status, so there may not be sufficient quantities available as ordered.
Some time may be required for production after orders.
Lead times may be longer depending on material acquisition status and modification processes.

*Note 1 Standard machined products: J Series, F Series, R Series, E Series, Hardened Plus, Hardened Plus J Series

Spur Gears

MSGA/MSGB Ground Spur Gears — Material: SCM415 — *m*1~4 Page 54
KSG Ground Spur Gears — Material: SCM440 — *m*1~3 Page 58
SSGS Ground Spur Pinion Shafts — Material: S45C — *m*1.5~3 Page 60
SSG Ground Spur Gears — Material: S45C — *m*0.5~10 Page 62
SSG F Series Ground Spur Gears — Material: S45C — *m*2~3 Page 80
SSG E Series Ground Spur Gears — Material: S45C — *m*1.5~6 Page 86
SSG R Series Ground Spur Gears — Material: S45C — *m*1.5~6 Page 96
Pinion Gears for Nabtesco GH Series — Material: S45C/SCM440 — *m*3~6 (CP10~20) Page 98
SSAG Ground Spur Gears — Material: S45C — *m*1~6 Page 100
KS Thermal Refined Spur Gears — Material: SCM440 — *m*1.5~5 Page 104
SSS Spur Pinion Shafts — Material: S45C — *m*1, 1.5 Page 106
SS-H Hardened Spur Gears — Material: S45C — *m*1~10 Page 110
SS Spur Gears — Material: S45C — *m*0.5~10 Page 108
SS F Series Spur Gears — Material: S45C — *m*1.5~3 Page 134
SSA-H Hardened Spur Gears — Material: S45C — *m*1~5 Page 142
SSA Spur Gears — Material: S45C — *m*1~5 Page 142
SSA F Series Spur Gears — Material: S45C — *m*2~3 Page 148
SSY-H Hardened Spur Gears — Material: S45C — *m*0.8, 1 Page 152
SSY Spur Gears — Material: S45C — *m*0.8, 1 Page 152
SSAY-H Hardened Spur Gears — Material: S45C — *m*1 Page 154
SSAY Spur Gears — Material: S45C — *m*1 Page 154
SUS/SUSA Stainless Steel Spur Gears — Material: SUS303 — *m*1~4 Page 156
SUSF F-Loc Gears — Material: SUS303 — *m*0.5, 1 Page 162
DSF F-Loc Gears — Material: Polyacetal (SUS303) — *m*0.5, 1 Page 164
NSU Plastic Spur Gears with Steel Core — Material: MC602ST (S45C) — *m*1~3 Page 166
PU Plastic Spur Gears with Steel Core — Material: MC901(SUS303) — *m*1~2 Page 170
PS/PSA Plastic Spur Gears — Material: MC901 — *m*1~3 Page 172
SUKB Stainless Steel Hubs — PSA Dedicated — Material: SUS303 — *φ*30~100 Page 182
PSUKB SUKB Assembled PSA Spur Gear — Material: MC901/SUS303 — *m*2~3 Page 183
DS Injection Molded Spur Gears — Material: Duracon (R)(M90-44) — *m*0.5~1 Page 184
BB Sintered Metal Bushings — Material: Oil-free copper alloy — *φ* 5~8 Page 186
BSS Spur Gears — Material: Free cutting brass (C3604) — *m*0.5~1 Page 188
SSR Steel Ring Gears (Spur Gears) — Material: S45C — *m*2~3 Page 190

M Includes Made to Order

Catalog Number of KHK Stock Gears

The Catalog Number for KHK stock gears is based on the simple formula listed below. Please order KHK gears by specifying the Catalog Numbers.

(Example) Spur Gears

S S G 1 - 15

- 15: No. of Teeth (15)
- 1: Module (1)
- G: Other Products (Ground Gears)
- S: Type (Spur Gear)
- S: Material (S45C)

Material	
M	SCM415
K	SCM440
S	S45C
SU	Stainless Steel
P	MC901
N	MC602ST
D	Polyacetal
BS	Brass
L	Sintered Metal Alloy

Type	
S	Spur Gears

Other Information	
A	Without Hub
G	Ground Gears
F	F-loc Hub Gears
R	Ring Gears
S	Pinion Shafts
U	Plastic Gears with Steel Core
Y	Thin Face Gears
H	Gear Teeth Induction Hardened

Features

To meet your applications, KHK stock gears are made in a variety of types, materials, configurations, modules and numbers of teeth. We also provide finished gears that are ready to use. Secondary operations can be performed to many of the products, allowing for a wider range of designs. The following table lists the main features.

Catalog Number	Module	Material	Heat Treatment	Tooth Surface Finish	Precision JIS B 1702-1:1998	Secondary Operations	Features
MSGA/MSGB	1~4	SCM415	Carburized	Ground	N5	×	Fully hardened, ground and keyway machined gears with excellent accuracy, strength and abrasion resistance.
KSG	1~3	SCM440	Thermal refined, gear teeth induction hardened	Ground	N6	△	Gears that have been tempered, hardened and ground that have excellent accuracy, strength and abrasion resistance. Secondary operations can be performed except for the teeth. This product is ideal for the pinion of the KRGF rack.
SSGS	1.5~3	S45C	Thermal refined, gear teeth induction hardened	Ground	N7	△	Gears with shafts that have been tempered, hardened and ground. Secondary operations can be performed except for the teeth.
SSG	0.5~10	S45C	Gear teeth induction hardened NOTE 1	Ground	N7	△	Gears that have been hardened and ground with a good balance of accuracy, wear resistance and cost. Secondary operations are possible except for the teeth.
SSAG	1~6						
KS	1.5~5	SCM440	Thermal refined	Cut	N8	○	Tempered gears with excellent bending strength. The teeth can be additionally hardened. This product is ideal for the pinion of the KRF rack.
SSS	1, 1.5	S45C	Thermal refined NOTE 2	Cut	N8 NOTE 3	○	Gears with a tempered shaft.
SS	0.5~10	S45C	—	Cut	N8 NOTE 3	○	Many lineups are available at a low price. The teeth can be additionally hardened.
SSA	1~5						
SSY/SSAY	0.8, 1	S45C	—	Cut	N8 NOTE 3	○	Gears with narrow teeth. Suitable for light loads.
SUS/SUSA	1~4	SUS303	—	Cut	N8	○	Stainless steel gears with rust resistance.
SUSF	0.5, 1	SUS303	—	Cut	N8 NOTE 3	×	Stainless steel gears with rust resistance. Locking Hub allows easy attachment.
DSF	0.5, 1	Polyacetal (SUS303)	—	Cut	N10 NOTE 3	×	Gears made of polyacetal. Locking Hub allows easy attachment.
NSU	1~3	MC602ST (S45C)	—	Cut	N9	○	Steel hubs are fused and fixed to reinforced nylon gears for secure fastening.
PU	1~2	MC901 (SUS303)	—	Cut	N9	○	Stainless steel hubs are fused and fixed to nylon gears for secure fastening.
PS/PSA	1~3	MC901	—	Cut	N9	○	Nylon gears can be used with no lubrication.
DS	0.5~1	Duracon (R) (M90-44) NOTE 4	—	Injection Molded	N12 equivalent	△	Low-priced gears made through injection molding. Suitable for light loads.
BSS	0.5~1	Free-cutting Brass (C3604)	—	Cut	N8 NOTE 3	○	Brass gears with excellent machinability.
SSR	2~3	S45C	—	Cut	N9	○	They have a ring shape with a large number of teeth.

○ Possible △ Partly possible × Not possible

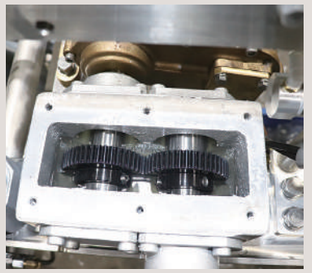
[NOTE 1] Products with module under 1 are thermal refined. Gear teeth are not hardened.
[NOTE 2] SA-shaped products with module 1 have no material thermal refinement treatment.
[NOTE 3] The product accuracy class having a module under 1 corresponds to 'equivalent' as shown in the table.
[NOTE 4] "Duracon (R)" is a registered trademark of Polyplastics Co., Ltd. in Japan as well as other countries.

● **KHK stock spur gears (m1.5 and higher) have semi-topping on the tooth tips.**
● **Black products are KHK stock gears that have an applied black oxide coating for rust resistance.**

Application Examples

Spur gears are widely used in a diverse range of equipment such as food machinery and industrial machines.

■ **Fish processing machine manufactured by TOYO SUISAN KIKAI CO.,LTD.**



SS spur gears used for filleting fish

■ **Carton former**



SS spur gears used in automatic carton formers



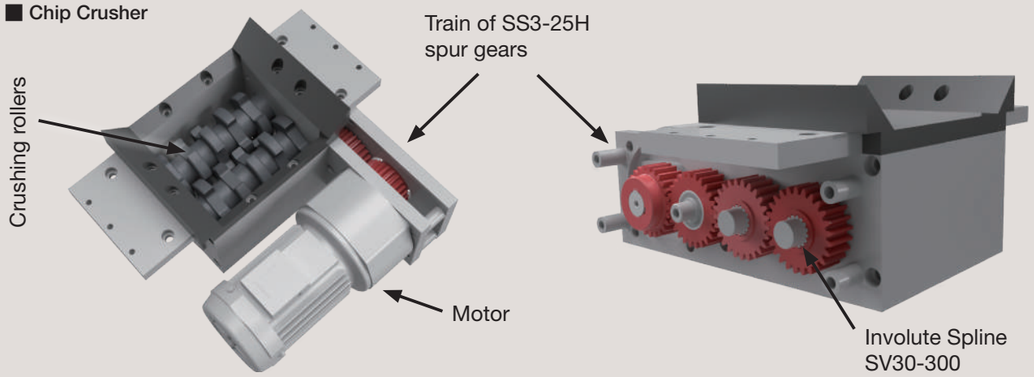
■ **High-speed automatic wire straightening/cutting machine manufactured by Takashima Sangyo Co.**



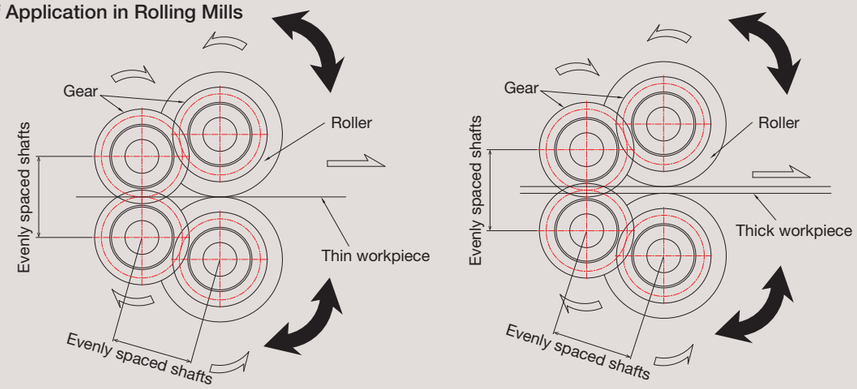
SS spur gears used for wire feeder



■ **Chip Crusher**



■ **Example of Application in Rolling Mills**



Selection Hints

Please select the most suitable products by carefully considering the characteristics of items and contents of the product tables. It is also important to read all applicable "CAUTION" notes shown below before the final selection.

1. Caution in Selecting the Mating Gears

① Basically, all spur gears, internal gears and racks can be paired as long as the module and pressure angle match. Products with different materials, tooth widths or accuracy can be mated.

2. Caution in Selecting Gears Based on Gear Strength

The gear strength values shown in the product pages were computed by assuming the application environment in the table below. Therefore, they should be used as reference only. We recommend that each user computes their own values by applying the actual usage conditions. Also, F-loc hub spur gears and various F series that use the friction coupling method to fasten the gear shaft need additional consideration for starting torque.

■ Calculation of Bending Strength of Gears

Item \ Catalog Number	MSGA MSGB	SSGS	SSG SSAG	SSS,SS SSA,SSY SSAY,SSR	SS-H SSY-H SSAY-H	SUS SUSA SUSF	BSS	KSG	KS	KS-H	ZSTP Note 6	SSG SSCPG Note 6	NSU	PU PS PSA	DSF DS
Formula NOTE 1	Formula of spur and helical gears on bending strength (JGMA401-01)												The Lewis formula		
No. of teeth of mating gears	Same number of teeth (30 for SSGS, SSS, SSR)							Racks					—		
Rotational Speed	600rpm NOTE 2			100rpm							400rpm		100rpm		
Design Life (Durability)	Over 10^7 cycles												—		
Impact from motor	Uniform load												Allowable bending stress (kgf/mm²)		
Impact from load	Uniform load												1.38 (40°C with no lubrication)	1.15 (40°C with no lubrication)	*m* 0.5 4.0 *m* 0.8 4.0 *m* 1.0 3.5 (40°C with grease lubrication)
Direction of load	Bidirectional load (calculated with allowable bending stress of 2/3)														
Allowable bending stress at root σ_{Flim} (kgf/mm²)	47	24.5	19 (24.5) Note 3	19 (24.5) Note 4	19	10.5	4	30	29.5	30	30	19			
Safety factor S_F	1.2														

■ Calculation of Surface Durability (Except where it is common with bending strength)

Item													
Formula NOTE 1	Formula of spur and helical gears on surface durability (JGMA402-01)												
Kinematic viscosity of lubricant	100cSt (50°C)												
Gear support	Symmetric support by bearings Note 5							Supported on one end.					
Allowable Hertz stress σ_{Hlim} (kgf/mm²)	166	99	90 (62.5) Note 3	49 (62.5) Note 4	90	41.3	—	112	76	112	112	90	
Safety factor S_H	1.15												

[NOTE 1] The gear strength formula is based on JGMA (Japanese Gear Manufacturers Association) specifications, "MC Nylon Technical Data" by Mitsubishi Chemical Advanced Materials and "Duracon (R) Gear" by Polyplastics Co. The units for the rotational speed (rpm) and the stress (kgf/mm²) are adjusted to the units needed in the formula.
[NOTE 2] For semi-custom gears, the rotation speed is based on 300rpm.
[NOTE 3] For SSG Ground Spur Gears, with module under 1, thermal refining is applied. Allowable bending stress and allowable hertz stress values are shown in parentheses.
[NOTE 4] For SSS Spur Pinion Shafts, with module over 1.5, tooth induction hardening is not applied. Allowable bending stress and allowable hertz stress values are shown in parentheses.
[NOTE 5] SSS Spur Pinion Shafts with module 1 or less (SA configuration) are set to cantilever support as they are single shaft types.
[NOTE 6] For Nabtesco GH Series.

When selecting KHK standard gears, glance over the Product Precautions on Page 50 and Cautions on Performing Secondary Operations on each page.

① Products not listed in this catalog or materials, modules, number of teeth and the like not listed in the dimensional tables can be manufactured as custom items. Please see Page 26 for more details.
② The color and shape of the product images listed on the dimension table page of each product may differ from the actual product.
Be sure to confirm the shape in the dimension table before selection.
③ The details (specifications, dimensions, etc.) listed in the catalog may be changed without prior notice. Changes are announced on the KHK website.

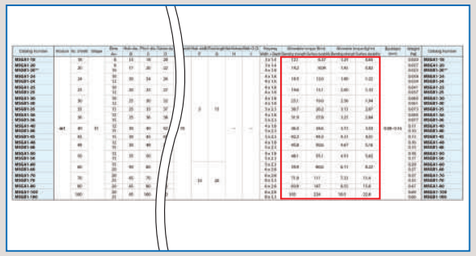
Website URL: https://khkgears.net/new/
Overseas Sales Department: Phone: +81-48-254-1744 Fax: +81-48-254-1765
E-mail: info@khkgears.net

Selecting the Gears

Step 1 Determine the calculated load torque applied to the gear and the gear type suitable for the purpose.

Step 2 Select provisionally from the allowable torque table in this catalog based on the load torque.

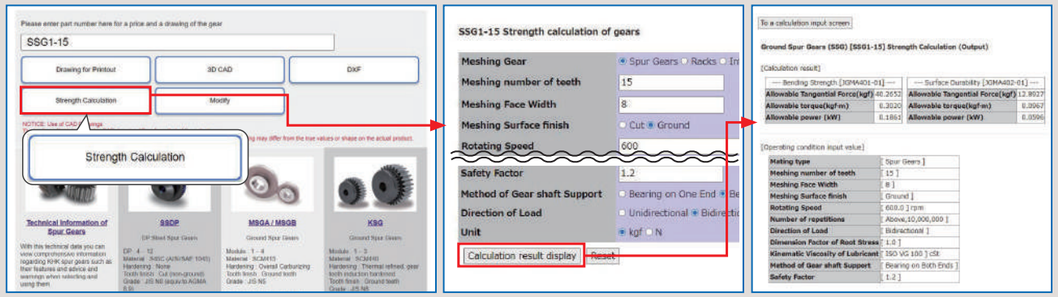
■ For provisional selection from this catalog



Step 3 Calculate the strength under the actual usage conditions.

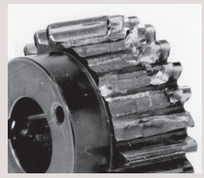
Calculate the strength formally using the various gear strength formulas. We recommend using the simple strength calculation available on our website.

■ Use the strength calculation function on our website.



■ Bending strength

Calculated values of the strength at which the gear teeth do not break due to fatigue.



Example of failure due to insufficient bending strength

■ Surface durability

Calculated values of the strength at which the gear teeth do not wear due to surface fatigue damage.



Example of wear due to insufficient surface durability

Product Precautions

Common Notes

[Caution on Product Characteristics]

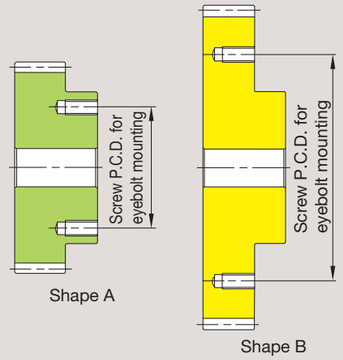
(1) The allowable torque shown in the table are calculated values according to the assumed usage conditions. Please see page 48 for more details.
(2) The backlash values shown in the table are the theoretical values for the backlash in the normal direction of a pair of identical gears in mesh.
(3) For hole lengths 3.5x the bore or more, the hole center is out of H7 tolerance.
(4) For bores of ϕ4 or below, the bore tolerance is H8. As well, the tolerance is H8 for ϕ5 or ϕ6 bores with hole length (total length) 3x the bore or more.
(5) Keyways are made according to JIS B1301 standards, Js9 tolerance. Also note that keyway tooth position alignment is not performed.
(6) For products having a tapped hole, a set screw is included.
(7) Variations in temperature or humidity can cause dimensional changes in plastic gears, including tooth diameter, bore, and backlash.
The accuracy and tolerances shown in the catalog are values obtained when machining is performed.
(8) To reduce heat generation, it is recommended to mate plastic gears with steel gears.
(9) See page 22 for more details on Hardened Plus (H Series and HJ Series).

● KHK's Specifications for Heat Treatment
Hardened location: Tooth surface, or Tooth surface and Tooth root
Hardness: 50 to 60 HRC
* Hardness and Depth of Gear-teeth Induction Hardening
The hardening method and the state of the hardened teeth area vary depending on the size of gears.
Since different hardening treatment is applied in accordance with the module and number of teeth, the hardness level is referred to as the hardness of the reference diameter.
For some of our products, the hardness at tooth tip / root may not be equal to the hardness you designated.
As to the effective case depth, it is specified by JIS, as "The distance from the surface of the case to the area with hardness HV450." The case depth differs from area to area of a tooth, so the depth cannot be specified.
Due to the gear teeth being induction hardened, no secondary operations can be performed on tooth areas including the bottom land (approx. 2 to 3 mm).

[J Series]

(1) Certain products which would otherwise have a very long tapped hole are counterbored. For details, please see the KHK website.
(2) Black oxide is not re-applied to parts undergoing secondary operations.
(3) For bores over ϕ50, the bore tolerance is H8.

Tapped Holes in Semi-Custom Standard Products



Catalog Number	Shape	Screw P.C.D. for eyebolt mounting
SSG4-62S	B	180
SSG4-64S	B	190
SSG4-65S	B	190
SSG4-66S	B	190
SSG4-68S	B	200
SSG4-70S	B	200
SSG4-80S	B	220
SSG4-90S	B	240
SSG4-100S	B	260
SSG4-110S	B	280
SSG4-120S	B	300
SSG5-32S	A	95
SSG5-35S	A	105
SSG5-36S	A	105
SSG5-40S	A	115
SSG5-42S	A	115
SSG5-44S	A	125
SSG5-45S	A	125
SSG5-48S	A	125
SSG5-50S	A	125
SSG5-55S	B	210
SSG5-56S	B	210
SSG5-60S	B	210
SSG5-62S	B	220
SSG5-64S	B	220
SSG5-65S	B	220
SSG5-66S	B	230
SSG5-68S	B	230
SSG5-70S	B	240
SSG5-80S	B	260
SSG5-90S	B	290
SSG6-32S	A	125
SSG6-35S	A	135
SSG6-36S	A	135
SSG6-40S	A	145
SSG6-42S	A	145
SSG6-44S	A	145
SSG6-45S	A	155
SSG6-48S	A	155
SSG6-50S	A	155
SSG6-55S	B	240
SSG6-56S	B	240
SSG6-60S	B	240
SSG6-62S	B	250
SSG6-64S	B	250
SSG6-65S	B	260
SSG6-66S	B	260

Catalog Number	Shape	Screw P.C.D. for eyebolt mounting
SSG6-68S	B	260
SSG6-70S	B	270
SSG6-80S	B	300
SSG8-20S	A	105
SSG8-25S	A	135
SSG8-30S	A	165
SSG8-32S	A	175
SSG8-35S	A	185
SSG8-36S	A	185
SSG8-40S	A	195
SSG8-42S	A	205
SSG8-44S	A	205
SSG8-45S	B	290
SSG8-48S	B	300
SSG8-50S	B	300
SSG8-55S	B	300
SSG8-56S	B	300
SSG8-60S	B	300
SSG10-20S	A	135
SSG10-25S	A	175
SSG10-30S	A	215
SSG10-32S	A	225
SSG10-35S	A	235
SSG10-36S	A	245
SSG10-40S	A	255

Catalog Number	Shape	Screw P.C.D. for eyebolt mounting
SS4-90S	B	240
SS4-100S	B	260
SS4-110S	B	280
SS4-120S	B	300
SS5-62S	B	220
SS5-64S	B	220
SS5-65S	B	220
SS5-66S	B	230
SS5-68S	B	230
SS5-70S	B	240
SS5-80S	B	260
SS5-90S	B	290
SS5-100S	B	310
SS5-110S	B	340
SS5-120S	B	360
SS6-52S	B	240
SS6-54S	B	240
SS6-55S	B	240
SS6-56S	B	240
SS6-58S	B	240
SS6-60S	B	240
SS6-62S	B	250
SS6-64S	B	250
SS6-65S	B	260
SS6-66S	B	260
SS6-68S	B	260
SS6-70S	B	270
SS6-80S	B	300
SS6-90S	B	330
SS6-100S	B	360
SS8-32S	A	175
SS8-34S	A	185
SS8-35S	A	185
SS8-36S	A	185
SS8-38S	A	195
SS8-40S	A	195
SS8-42S	A	205
SS8-44S	A	205
SS8-45S	B	290
SS8-46S	B	215
SS8-48S	B	300
SS8-50S	B	300
SS8-52S	B	300
SS8-54S	B	300
SS8-55S	B	300
SS8-56S	B	300
SS8-58S	B	300

Catalog Number	Shape	Screw P.C.D. for eyebolt mounting
SS8-59S	B	300
SS8-60S	B	300
SS10-26S	A	185
SS10-27S	A	195
SS10-28S	A	195
SS10-29S	A	205
SS10-30S	A	215
SS10-32S	A	225
SS10-34S	A	235
SS10-35S	A	235
SS10-36S	A	245
SS10-38S	A	245
SS10-40S	A	255
SS10-42S	A	265
SS10-44S	B	350
SS10-45S	B	350
SS10-46S	B	350
SS10-48S	B	360
SS10-50S	B	360

Application Hints

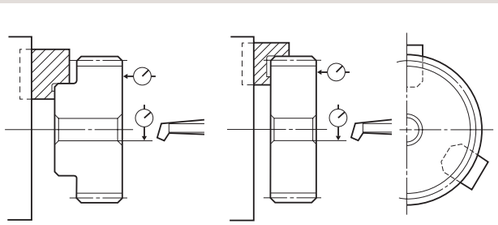
In order to use KHK stock gears safely, carefully read the Application Hints before proceeding.
If there are questions or you require clarifications, please contact your nearest distributor. E-mail: info@khkgears.net
Please read "Cautions on Performing Secondary Operations" below when performing modifications and/or secondary operations for safety concerns.

1. Cautions on Handling

① KHK products are packaged one by one to prevent scratches and dents, but if you find issues such as rust, scratches, or dents when the product is removed from the box after purchase, please contact the supplier.
② Depending on the handling method, the product may become deformed or damaged. Plastic gears and ring gears deform particularly easily, so please handle with care.

2. Caution on Performing Secondary Operations

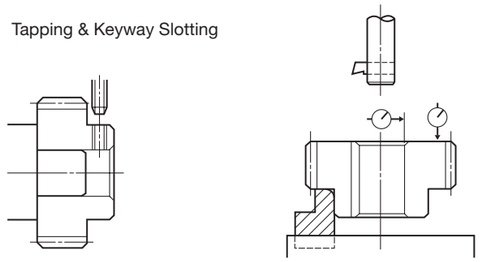
① If reboring, it is important to pay special attention to locating the center in order to avoid runout.
② The reference datum for gear machining is the bore. Therefore, use the bore for locating the center. If it is too difficult to do for small bores, the alternative is to use one spot on the bore and the runout of the side surface.
③ If reworking using scroll chucks, we recommend the use of new or rebored jaws for improved precision. Please exercise caution not to crush the teeth.



Lathe Operations

④ The maximum bore size is dictated by the requirement that the strength of the hub is to be higher than that of the gear teeth. The maximum bore size should be 60% to 70% of the hub diameter (or tooth root diameter), and 50% to 60% for keyway applied modifications.
⑤ In order to avoid stress concentration, round the keyway corners.

Tapping & Keyway Slotting



⑥ To avoid problems of reduced gear precision and other manufacturing difficulties, do not attempt to machine the gears to reduce face widths.
⑦ When induction-hardening S45C products, thermal stress cracks may appear. Also, note that the precision grade of the product declines by 1 or 2 grades, as deformation on material may occur. If you require tolerance for bore or other parts, machining is necessary after heat treatment.

Induction Hardening

If you apply induction hardening to the gear teeth of S45C products, you need to designate the hardness and where to apply the heat treatment. Below is an example of common specifications and KHK's specifications for hardening:

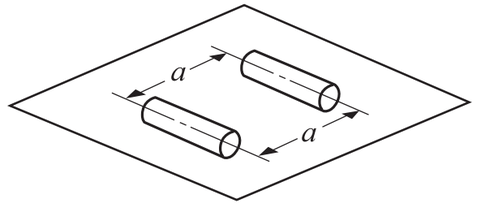
● Common Specifications for Heat Treatment
Hardened location: Tooth surface, or Tooth surface and Tooth root
Hardness: Within 10 HRC in the range from 45 to 60 HRC
(Example: 48 to 58 HRC)

● KHK's Specifications for Heat Treatment
Hardened location: Tooth surface, or Tooth surface and Tooth root
Hardness: 50 to 60 HRC

* Hardness and Depth of Gear-teeth Induction Hardening
The hardening method and the state of the hardened teeth area vary depending on the size of gears.
Since different hardening treatment is applied in accordance with the module and number of teeth, the hardness level you designate is referred to as the hardness of the reference diameter. For some of our products, the hardness at tooth tip / root may not be equal to the hardness you designated.
As to the effective case depth for S45C, it is specified by JIS, as "The distance from the surface of the case to the area with hardness HV450." The case depth differs from area to area of a tooth, so the depth cannot be specified.

3. Points of Caution during Assembly

① The recommended center distance tolerance of KHK stock spur gears is H7 for ground gears and H8 for cut gears. Backlash may be adjusted by changing the center distance of mating gears. For the connection between center distance change amount and peripheral direction backlash amount, use the gear calculation software.



$a=m(Z_1+Z_2)/2$

Where
a : Center distance
m : Module
Z_1 : No. of teeth of pinion
Z_2 : No. of teeth of gear

② The table below indicates the tolerance on the total length of KHK stock spur gears. Please refer to this data when designing gearboxes or other components.

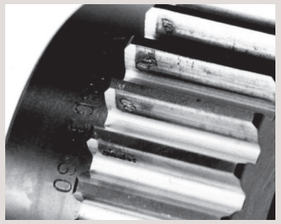
■ Total Length Tolerance for Spur and Helical Gears

Total Length (mm)	Tolerance
30 or less	0 -0.10
31 to 100	0 -0.15
Over 100	0 -0.20

[Note] The following products are excluded from this table: Hardened Plus, Spur pinion shafts, Injection molded spur gears, F-loc hub spur gears, and MC nylon products.

③ Spur gears produce no thrust forces; however, be sure to fasten them firmly with stepped shafts, or collars, to prevent shifting toward the shaft.
Keyways are generally used in fastening gears to a shaft, and they should be fastened by applying drilled holes for set screws, or applying flats to the shaft, in case of fastening only with set screws.
There are also methods of secure settings using parts for engaging the hole and the axis.
④ Verify that the two shafts are parallel. Incorrect assembly will lead to uneven teeth contact which will cause noise and wear. (After assembly, check the tooth contact by painting a thin layer of red lead primer or the like on the gear teeth, meshing them together and rotating them.)

■ Test example: Abrasion occurred on SSG3-30 due to poor edge contact (only 30% with proper contact).



Poor tooth contact and pitting

In this example, the gear oil used is equivalent to the JIS gear oil category 2, No. 3
The design conditions were load torque at 278 rpm, 42.5 kg/m (12 kW), 1.5 times the allowable bending strength, and 3 times the allowable surface durability torque.
The pitting occurred on the poor tooth contact area after 60 hours of continuous operation.

4. Cautions on Starting

① Check the following items before starting.
- Are the gears fastened securely?
- Is there uneven tooth contact?
- Is there adequate backlash?
  (Be sure to avoid zero-backlash.)
- Has proper lubrication been supplied?

② If gears are exposed, be sure to attach a safety cover to ensure safety. Also, be careful not to touch rotating gears.
③ If there is any abnormality such as noise or vibration during startup, stop the operation immediately and check the assembly condition such as tooth contact, eccentricity and looseness.

KHK considers safety a priority in the use of our products.
When handling, adding secondary operations, assembling, and operating KHK products, please be aware of the following issues in order to prevent accidents.

Warning: Precautions for preventing physical and property damage

1. When using KHK products, follow relevant safety regulations (Occupational Safety and Health Regulations, etc.).
2. Pay attention to the following items when installing, removing, or performing maintenance and inspection of the product.
   ① Turn off the power switch.
   ② Do not reach or crawl under the product.
   ③ Wear appropriate clothing and protective equipment for the work.

Caution: Cautions in preventing accidents

1. Before using a KHK product, read the precautions in the catalog carefully in order to use it correctly.
2. Avoid use in environments that may adversely affect the product.
3. Our products are manufactured under a superior quality control system based on the ISO9001 quality management system; if you notice any malfunctions upon purchasing a product, please contact the supplier.